

Unsupervised Machine Learning arXiv Topic Clustering Report

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Goal

The goal of this homework is to solve a clustering-focused unsupervised learning problem on recent arXiv papers from `cs.LG` and `cs.CL`. I use paper titles and abstracts to discover broad research topics, inspect how those topics vary over time, and demonstrate a small content-based recommender.

The dominant assignment component is clustering. The recommender is included as a secondary use of the same document representation.

Course Alignment

The implementation follows the structure and tools covered in the course materials:

- **Document similarity:** cosine similarity is used for text vectors because vector direction is more informative than raw vector length for abstracts.
- **Dimensionality reduction:** TF-IDF is compressed with truncated SVD into latent semantic analysis (LSA) features, and PCA is used for a two-dimensional visualisation.
- **Clustering:** KMeans, Ward agglomerative clustering, and HDBSCAN are compared.
- **Internal validation:** Silhouette, Davies-Bouldin, and Calinski-Harabasz scores are reported for the clustering comparison.
- **Recommendation:** the final helper is content-based rather than collaborative because no user-paper rating matrix is available.

Method

1. **Data acquisition:** I fetched a balanced monthly sample of arXiv papers from the previous 12 months. The final cached corpus contains 1,848 papers, with 154 papers per month from May 2025 through April 2026.
2. **Text cleaning and representation:** Titles and abstracts are concatenated, URL-like tokens are removed, and a TF-IDF matrix is fitted with unigrams and bigrams. I also compute a 100-dimensional dense LSA representation with truncated SVD.
3. **Clustering comparison:** KMeans, Ward agglomerative clustering, and HDBSCAN are run on the dense LSA representation. The same algorithms are also evaluated on sparse TF-IDF, with dense conversion only where required by the implementation.
4. **Topic interpretation:** The best practical clustering result is interpreted with high-weight TF-IDF terms and representative papers closest to each KMeans centroid.

5. **Visualisation and recommendation:** I generate a PCA scatter plot, a topic-size bar chart, a smoothed topic timeline, and a content-based recommender using cosine similarity in LSA space.

Implementation Note

The initial plan included transformer-based scientific embeddings and a BERTopic layer. During execution, that path pulled a very large PyTorch/CUDA dependency chain and the install did not complete in a useful time window. Because the assignment is time-sensitive and clustering is the dominant requirement, I changed to a lighter and fully reproducible TF-IDF plus LSA pipeline. This still covers the planned clustering comparison, internal metrics, topic interpretation, visualisation, and recommender components.

Results

Representation	Method	Clusters	Outliers	Sil.	DB	CH
LSA	KMeans	14	0.0%	0.075	4.09	23.48
LSA	Ward	14	0.0%	0.029	4.50	17.77
LSA	HDBSCAN	4	91.2%	0.257	2.33	17.01
TF-IDF	KMeans	16	0.0%	0.007	10.27	3.32
TF-IDF	Ward	16	0.0%	0.002	9.31	3.19
TF-IDF	HDBSCAN	0	100.0%	–	–	–

Table 1: Internal clustering metrics. HDBSCAN scores are computed on non-outlier papers only, so its high LSA silhouette is not directly comparable to the full-coverage KMeans result.

KMeans on cleaned LSA is the best practical configuration. HDBSCAN identifies small dense groups, but it marks most papers as outliers. Direct TF-IDF clustering is much weaker by all internal metrics.

Topic	Keyword label	Papers
5	llms, language, language models, llm	254
3	training, model, models, data	254
0	data, learning, optimization, gradient	223
13	machine, machine learning, learning, data	173
4	neural, networks, neural networks, graph	163
1	agent, agents, multi, multi agent	135
8	diffusion, generative, generation, diffusion models	103
10	reasoning, llms, cot, models	94

Table 2: Largest KMeans-LSA topics after cleaning URL-like tokens.

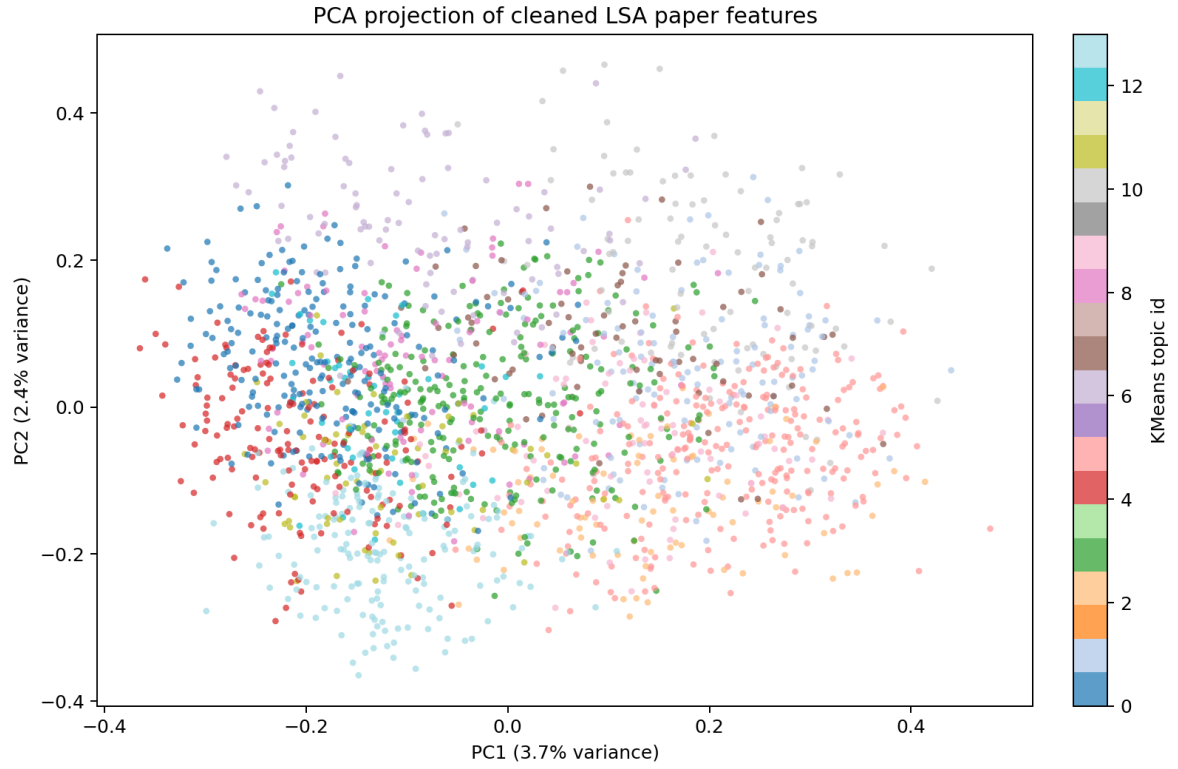


Figure 1: PCA projection of the cleaned LSA features, coloured by KMeans topic id. The overlap supports the interpretation that arXiv topics are continuous and partially mixed rather than sharply separated.

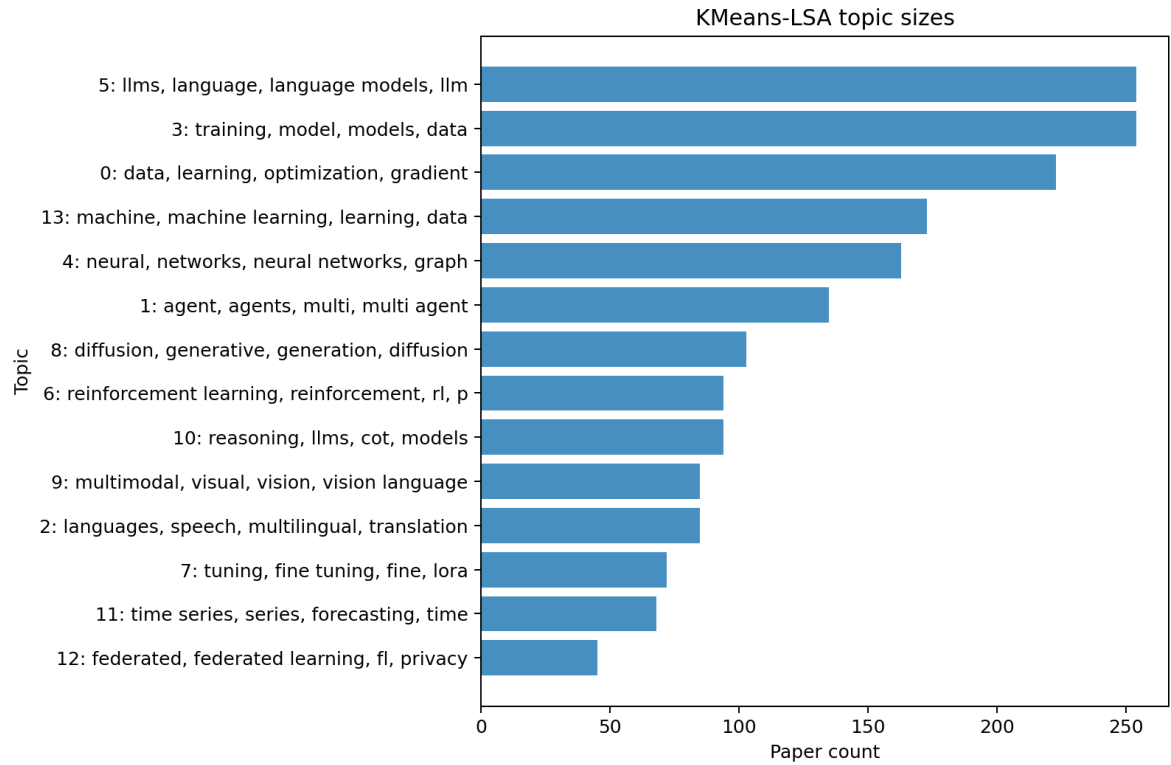


Figure 2: Topic sizes for the selected KMeans-LSA model.

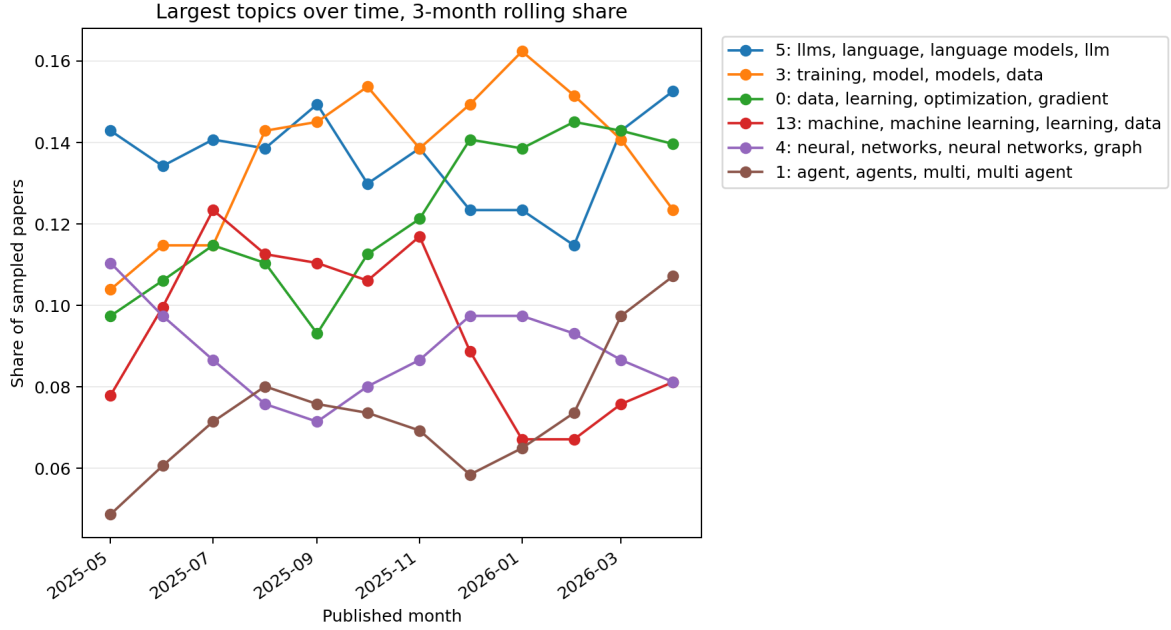


Figure 3: Three-month rolling topic share for the largest topics. The plot is exploratory because each month contains a fixed sample of about 154 papers.

Discussion

The KMeans elbow curve is fairly gradual, which is expected for research abstracts. Papers often combine methods and application areas, so the corpus does not form perfectly separated geometric clusters. I therefore treat the selected number of topics as a useful granularity, not as a unique natural truth.

The strongest result is the representation comparison: raw sparse TF-IDF is not enough for coherent clustering, while cleaned LSA gives a much more useful topic structure. The URL-token cleaning step also mattered: without it, one cluster was dominated by repository-link tokens rather than scientific content.

The recommender demonstrates a second use of the learned representation. Given a query abstract, the system transforms it with the fitted TF-IDF and LSA models, then returns papers with the highest cosine similarity. This is a reasonable content-based alternative when collaborative filtering data is not available.

Conclusion

The final notebook completes a clustering-dominant unsupervised learning pipeline with data acquisition, representation learning, algorithm comparison, topic interpretation, visualisation, and content-based recommendation. The main practical conclusion is that cleaned TF-IDF compressed with LSA and clustered by KMeans is a simple, reproducible baseline for exploring recent arXiv topics under tight runtime constraints.